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TOPIC 1: What Is Risk?

Core Definition

According to Pressman, every software risk has exactly two essential characteristics. Both must be present for something to qualify as a risk.

Characteristic	Meaning
Uncertainty	The risk may or may not happen. There are no 100% probable risks. If something is 100% certain, it is a constraint — not a risk.
Loss	If the risk becomes real, unwanted consequences will occur — such as schedule delays, cost overruns, or product failure.

■ *Analogy: Imagine you are launching a mobile app. There is a chance your lead developer might quit mid-project. That chance of quitting is the uncertainty. If they do quit, your timeline and cost blow up — that is the loss. Together, that is a risk.*

Risk vs Problem vs Constraint

This distinction is commonly tested in exams:

Term	Meaning
Risk	A potential future problem — has not happened yet
Problem / Issue	A risk that has already become real
Constraint	A certainty that limits the project — not a risk at all

Peter Drucker quote used by Pressman: "While it is futile to try to eliminate risk, and questionable to try to minimize it, it is essential that the risks taken be the right risks."

Key Exam Points

- ✓ A risk always involves BOTH uncertainty AND loss — both must be present.
- ✓ A 100% probable event is a CONSTRAINT, not a risk.
- ✓ Risk management is about identifying the RIGHT risks — elimination of all risk is impossible.
- ✓ Once a risk becomes real, it becomes a problem/issue — no longer a risk.

Exam Questions & Model Answers

■ **Exam Q: Define software risk. What are its two essential characteristics?**

Model Answer: A software risk is any potential event or condition that threatens the success of a software project. Every risk has two essential characteristics: (1) Uncertainty — the risk may or may not occur, meaning it is not a certainty; and (2) Loss — if the risk does occur, it will result in unwanted consequences such as schedule delays, cost overruns, or product failure. Both characteristics must be present for something to qualify as a risk. An event that is 100% certain is not a risk but a project constraint.

TOPIC 2: Reactive vs Proactive Risk Strategies

The Two Philosophies

Every software team has a philosophy about how to deal with risks. Pressman presents two opposing approaches:

Philosophy	Attitude	Nickname
Reactive	We will deal with it when it happens	The Indiana Jones approach
Proactive	We plan before problems arrive	The intelligent approach

Reactive Strategy — The Indiana Jones School

Pressman uses the Indiana Jones analogy: Indy never worried about problems until they happened — then he heroically reacted. Most software teams operate exactly like this. A reactive strategy has three levels, from bad to worse:

Level	Name	Description
Level 1	Monitor Only	Team watches for risks but does nothing until the risk strikes. Resources set aside just in case. Best form of reactive behavior.
Level 2	Fix on Failure	Nothing is done until the risk actually hits. Team scrambles, throws resources at the problem, tries to recover. Most common reactive approach.
Level 3	Crisis Management	Project has already gone seriously wrong. Team is trying to survive, not succeed. Pressman says: the project is in real jeopardy. Worst outcome.

■ *Example: A team builds an e-commerce site. Nobody plans for traffic spikes. Launch day arrives, a million users hit the site, servers crash. The team is now in crisis mode — fixing something that was entirely predictable.*

Proactive Strategy — The Intelligent Approach

A proactive strategy begins long before technical work starts. The sequence:

1. Identify risks early
2. Assess probability and impact of each risk
3. Rank risks by importance
4. Create a plan to avoid or handle each risk
5. Build contingency plans for risks that cannot be avoided

6. Monitor throughout the entire project lifecycle

■ *Analogy: A pilot does a full pre-flight checklist before every flight — not because something has gone wrong, but because something might. That is proactive risk management.*

The 7 Principles of Proactive Risk Management (SEI)

The Software Engineering Institute (SEI) defines 7 principles for effective risk management:

Principle	What It Really Means
Maintain a global perspective	Do not look at risks in isolation. A risk in one module can cascade across the whole system. See the big picture.
Take a forward-looking view	Think about risks that will arise as the project evolves. Build contingency plans now, not later.
Encourage open communication	If a junior developer says 'I think this might be a problem' — listen. Do not dismiss concerns. Psychological safety is essential.
Integrate	Risk management is not a one-time activity. It must be woven into every phase of software development.
Emphasize a continuous process	Risks evolve. New risks appear. Old ones disappear. Keep monitoring throughout the entire project lifecycle.
Develop a shared product vision	When all stakeholders agree on what is being built, risk identification becomes more accurate and everyone stays aligned.
Encourage teamwork	Pool the knowledge of everyone — developers, testers, managers, clients. Different people see different risks.

Memory Trick for 7 Principles: G-F-O-I-C-S-T — "Good Firms Often Integrate Continuous Shared Teamwork"

Key Exam Points

- ✓ Reactive strategy = fire-fighting; Proactive strategy = fire-prevention.
- ✓ Three levels of reactive: Monitor Only → Fix on Failure → Crisis Management.
- ✓ Proactive strategy starts BEFORE technical work begins.
- ✓ The primary objective of proactive strategy is RISK AVOIDANCE.
- ✓ The SEI defines 7 principles for effective risk management.
- ✓ Crisis management is the WORST possible outcome of reactive thinking.

Exam Questions & Model Answers

■ **Exam Q:** Differentiate between reactive and proactive risk strategies with examples.

Model Answer: A reactive risk strategy means the team takes no action until a risk becomes an actual problem. At best, resources are set aside to deal with problems when they occur; at worst, the project enters crisis management mode where failure is imminent. For example, a team that discovers a missing security module only after deployment is being reactive. A proactive risk strategy begins identifying and assessing risks before technical work starts. Risks are ranked by probability and impact, and mitigation plans are developed in advance. For example, identifying that a third-party API might be unreliable and building a fallback mechanism before development begins is proactive risk management. Pressman strongly advocates the proactive approach as it leads to more controlled, predictable projects.

■ **Exam Q: What is crisis management in the context of software risk?**

Model Answer: Crisis management is the most severe form of reactive risk strategy. It occurs when a risk has become a reality, mitigation has failed, and the project is in serious jeopardy of complete failure. At this stage, the team is no longer managing the project normally — they are simply trying to survive. Pressman notes that crisis management represents a failure of risk planning and is the outcome of treating risk management as an afterthought rather than an integral part of software project planning.

TOPIC 3: Risk Identification (Section 28.3)

What Is Risk Identification?

Risk identification is a systematic attempt to specify threats to the project plan. The goal is to uncover known and predictable risks so the project manager can either avoid them or control them. There are two types of risks to identify:

Type	Meaning
Generic risks	Potential threats to EVERY software project — common across the industry. Well-known and manageable.
Product-specific risks	Risks unique to THIS particular project. Require deep understanding of the technology, people, and environment. These cause the most headaches.

Pressman's important note: "It is the product-specific risks that cause the most headaches. Be certain to spend the time to identify as many product-specific risks as possible."

The Risk Item Checklist — 7 Categories

The main tool for risk identification. Covers seven generic subcategories. Know all seven by name for the exam.

#	Category	What It Covers	Example
1	Product Size	Risks tied to the overall size of the software being built or modified.	A 500K LOC project has far more risk than a 5K LOC project.
2	Business Impact	Risks from management and marketplace constraints: deadlines, regulations, revenue impact.	Software controlling medical devices — a defect costs lives and legal liability.
3	Stakeholder Characteristics (NOT 'Customer' — slides have wrong term)	Risks from poor communication or inexperienced stakeholders.	A client who says 'just make it look nice' with no technical requirements.
4	Process Definition	Risks from an undefined or unfollowed software process.	A team with no review process or coding standards — like construction with no blueprint.

#	Category	What It Covers	Example
5	Development Environment	Risks from poor or unavailable tools.	Forced to use outdated compilers or IDEs with no debugging support.
6	Technology to be Built	Risks from complexity and newness of the technology.	A traditional company suddenly building AI-powered systems.
7	Staff Size and Experience	Risks from inadequate or inexperienced team members.	Five fresh graduates building a real-time banking system with no senior oversight.

Slide Error to Note: Your slide (Slide 7) says 'Customer characteristics.' The Pressman book uses 'Stakeholder characteristics' — a broader term that includes customers, managers, end-users, and developers.

Risk Category Abbreviations (from Pressman's Risk Table)

Pressman uses shorthand codes in his sample risk table (Figure 28.2). These map directly to the 7 categories above:

Code	Full Category Name	Note
PS	Product Size	Appears in sample table
BU	Business Impact	Appears in sample table
CU	Customer / Stakeholder	Appears in sample table
TE	Technology to be Built	Appears in sample table
DE	Development Environment	Appears in sample table
ST	Staff Size and Experience	Appears in sample table
PD	Process Definition	Valid 7th category — not in sample table but exists in checklists

Student Question: Where are these abbreviations listed and are there more? These codes come from Pressman's own Figure 28.2 example table — not a formal industry standard. They map 1-to-1 with the 7 risk identification categories. PD (Process Definition) is valid but does not appear in the sample table. In exams, use the full category name, not just the abbreviation.

Assessing Overall Project Risk — The 11 Questions (Section 28.3.1)

After identifying risks, you assess the overall health of the project using 11 key questions. Any 'No' answer is a red flag. The more No answers, the higher the overall risk. If ANY one is answered negatively, mitigation steps must be instituted immediately.

#	Question
1	Have top software and customer managers formally committed to support the project?
2	Are end-users enthusiastically committed to the project and the system to be built?
3	Are requirements fully understood by the software engineering team AND its customers?
4	Have customers been involved fully in the definition of requirements?
5	Do end-users have realistic expectations?
6	Is the project scope stable?
7	Does the software engineering team have the right mix of skills?
8	Are project requirements stable?
9	Does the project team have experience with the technology to be implemented?
10	Is the number of people on the project team adequate to do the job?
11	Do all customer/user constituencies agree on the importance of the project and on requirements?

Key Exam Points

- ✓ Two types of risks: GENERIC (common to all projects) and PRODUCT-SPECIFIC (unique to this project).
- ✓ Product-specific risks are more dangerous and harder to identify.
- ✓ The Risk Item Checklist covers 7 categories — know all 7 by name.
- ✓ 'Customer characteristics' in slides = 'STAKEHOLDER characteristics' in the book.
- ✓ The 11 questions assess overall project risk — any No answer = red flag.
- ✓ Risk identification is a SYSTEMATIC, not casual activity.

Exam Questions & Model Answers

■ Exam Q: List and briefly explain the risk categories used in the risk item checklist.

Model Answer: The risk item checklist identifies risks across seven generic subcategories: (1) Product size — risks from the overall size and complexity of the software; (2) Business impact — risks from marketplace and management constraints; (3) Stakeholder characteristics — risks from poor communication or inexperienced stakeholders; (4) Process definition — risks from an undefined or unfollowed software process; (5) Development environment — risks from poor or unavailable tools; (6) Technology to be built — risks from new or complex technology; and (7) Staff size and experience — risks from inadequate or inexperienced team members. These categories help project managers systematically uncover both known and predictable risks before technical work begins.

TOPIC 4: Risk Projection / Risk Estimation (Section 28.4)

What Is Risk Projection?

Risk projection (also called risk estimation) attempts to rate each identified risk in two ways: (1) Probability — how likely is it that this risk will actually occur? and (2) Consequences/Impact — how bad will it be if it does occur? The goal is PRIORITIZATION — you cannot manage every risk equally.

The 4 Steps of Risk Projection

Step 1: Establish a scale that reflects the perceived likelihood of a risk

Step 2: Delineate the consequences of the risk

Step 3: Estimate the impact on the project and the product

Step 4: Assess accuracy of the projection so there are no misunderstandings

Risk Components and Their Impact Categories (Section 28.3.2)

Slide gap: Your slides list the 4 risk components but completely miss the impact categories.

The 4 Risk Components

Component	What It Measures
Performance risk	Will the product meet its requirements and be fit for use?
Cost risk	Will the project budget be maintained?
Support risk	Will the software be easy to correct, adapt, and enhance?
Schedule risk	Will the project be delivered on time?

The 4 Impact Categories

Category	What It Means
Catastrophic	Mission failure — the system simply cannot work. Unachievable goals.
Critical	Significant degradation — serious budget overruns or major delays.
Marginal	Minor reduction — small, recoverable problems.

Category	What It Means
Negligible	No real impact — inconvenience only.

Memory Trick for impact categories: "Can Cats Meet Naps" — Catastrophic, Critical, Marginal, Negligible

Developing a Risk Table (Section 28.4.1)

The Risk Table is the central tool of risk projection — a spreadsheet organizing all risks so you can prioritize them clearly.

How to Build the Risk Table — Step by Step

Step 1 — List ALL risks: No matter how unlikely or unusual. Do not filter yet. Cast a wide net.

Step 2 — Estimate probability: Each team member independently estimates probability, then the group converges on a consensus value through round-robin discussion. Scale: Impossible → Improbable → Probable → Frequent.

Step 3 — Assess impact: Rate each risk using the 4 impact categories. Scale used: 1=Catastrophic, 2=Critical, 3=Marginal, 4=Negligible.

Step 4 — Sort the table: Sort by probability first, then impact. High-probability + high-impact risks rise to the top. This is called FIRST-ORDER PRIORITIZATION.

Step 5 — Draw the cutoff line: A horizontal line separates risks that will be actively managed (above) from those requiring only secondary analysis (below). Only risks above the cutoff get an RMMM entry.

Sample Risk Table (based on Pressman's Figure 28.2)

Note: Your slides show 4 columns in the risk table. The category column below comes from the book's Figure 28.2.

Risk Description	Category	Probability	Impact	RMMM
Customer will change requirements	PS	80%	2 – Critical	See Plan #1
Lack of training on tools	DE	80%	3 – Marginal	See Plan #2
Less reuse than planned	PS	70%	2 – Critical	See Plan #3
Staff turnover will be high	ST	60%	2 – Critical	See Plan #4
Delivery deadline tightened	BU	50%	2 – Critical	See Plan #5
End-users resist system	BU	40%	3 – Marginal	See Plan #6
Funding will be lost	CU	40%	1 – Catastrophic	See Plan #7

■■■ CUTOFF LINE ■■■	■■■	■■■	■■■	Below: reevaluate
Technology will not meet expectations	TE	30%	1 – Catastrophic	Monitor only
Staff inexperienced	ST	30%	2 – Critical	Monitor only
Larger number of users than planned	PS	30%	3 – Marginal	Monitor only

Table sorted by probability then impact. Yellow row = cutoff line.

Management Concern vs Probability and Impact

Pressman's Figure 28.3 shows how probability and impact together drive management concern:

Situation	Management Action
High impact + Very low probability	Do NOT waste significant management time on it
High impact + Moderate to high probability	MUST be actively managed — appears above cutoff line
Low impact + High probability	Should still be carried forward for analysis

■ *Analogy: A meteor hitting your office has catastrophic impact but near-zero probability — you do not design your office around it. But a power outage has moderate impact and reasonable probability — you buy a UPS backup. That is how management concern works.*

The Risk Exposure Formula (Section 28.4.2)

The Risk Exposure (RE) formula quantifies the financial impact of a risk:

$$RE = P \times C$$

Symbol	Meaning
RE	Risk Exposure — total financial impact of the risk
P	Probability of occurrence (as a decimal, e.g. 0.80 for 80%)
C	Cost to the project if the risk actually occurs (in dollars)

Worked Example from Pressman

Risk identified: Only 70% of software components scheduled for reuse will actually be integrated. The remaining 30% must be custom developed.

Step	Calculation	Result
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Risk probability	Given	80% = 0.80
Components planned for reuse	Given	60 components
Components actually usable (70%)	60×0.70	42 components
Components to build from scratch	$60 - 42$	18 components
Average component size	Given	100 LOC
Cost per LOC	Given	\$14.00
Total cost of custom development	$18 \times 100 \times \$14$	\$25,200
RISK EXPOSURE (RE)	$0.80 \times \\$25,200$	$\approx \\$20,200$

RE can be computed for every risk above the cutoff line. The total RE across all risks can be added to the project cost estimate as a financial buffer. If total RE exceeds 50% of project cost, the viability of the project must be reconsidered.

Three Factors That Affect Risk Impact

When assessing consequences, three factors determine the severity:

Factor	What It Means
Nature	What kind of problems will occur if the risk hits?
Scope	How severe is it, and how much of the project does it affect?
Timing	When will the impact be felt — early or late in the project? Bad news early is better than bad news late.

Key Exam Points

- ✓ Risk projection rates each risk on PROBABILITY and CONSEQUENCES — both.
- ✓ 4 risk components: Performance, Cost, Support, Schedule — remember 'PCSS'.
- ✓ 4 impact categories: Catastrophic, Critical, Marginal, Negligible — remember 'CCMN'.
- ✓ Risk Table is sorted by probability then impact = FIRST-ORDER PRIORITIZATION.
- ✓ The CUTOFF LINE separates actively managed risks from reevaluated ones.
- ✓ $RE = P \times C$ — know the formula and be able to apply it with calculations.
- ✓ Three factors affecting impact: Nature, Scope, Timing.
- ✓ Only risks ABOVE the cutoff line get an RMMM column entry.

Exam Questions & Model Answers

■ **Exam Q: What is Risk Exposure? How is it calculated? Give an example.**

Model Answer: Risk Exposure (RE) is a measure that quantifies the overall impact of a risk on a project in financial terms. It is calculated using the formula $RE = P \times C$, where P is the probability of the risk occurring and C is the cost to the project if the risk does occur. For example, if there is an 80% probability that 18 software components will need to be custom developed at a cost of \$25,200, then $RE = 0.80 \times \$25,200 = \$20,160$ (approximately \$20,200). This figure can be used to adjust the overall project cost estimate and helps managers prioritize which risks deserve the most resources and attention.

■ **Exam Q: What is a risk table and how is it used in risk projection?**

Model Answer: A risk table is a systematic tool used during risk projection to list, categorize, and prioritize all identified project risks. Each row contains a risk description, its category, its probability of occurrence, and its impact on the project. Once all risks are entered, the table is sorted by probability and impact so that high-probability, high-impact risks appear at the top — this is first-order prioritization. A cutoff line is then drawn to separate risks that will be actively managed from those requiring only secondary analysis. The final RMMM column points to the risk mitigation, monitoring and management plan for each risk above the cutoff line.

TOPIC 5: Risk Mitigation, Monitoring and Management (RMMM)

The RMMM Column in the Risk Table

Student Question: What goes in the 4th column of the risk table titled 'RMMM'? The RMMM column contains a POINTER or REFERENCE to the risk's mitigation, monitoring and management plan. In practice: a reference number pointing to a detailed plan document (e.g. 'See RMMM Plan #P02-4-32') or a brief one-line note summarizing the planned response. You do NOT write the entire plan inside the table cell — just a reference to where it lives. Only risks ABOVE the cutoff line get an RMMM entry.

The Three Components of RMMM

Together, these three activities form a complete risk response strategy covering before, during, and after a risk occurs.

Phase	Component	Question It Answers	When It Happens
BEFORE	Mitigation	How can we AVOID the risk?	Before the risk occurs
DURING	Monitoring	How do we TRACK if it gets worse?	Continuously throughout project
IF IT HITS	Management	What is our CONTINGENCY plan?	After risk becomes reality

1. Mitigation — How can we AVOID the risk?

Mitigation is about PREVENTION. You act before the risk becomes real. The goal is to reduce either the probability of the risk occurring or its impact if it does.

■ *Analogy: Mitigation is like wearing a seatbelt. You do it before the accident, not during.*

Pressman's example — High Staff Turnover Risk:

- Meet with current staff to find out WHY people might leave
- Fix whatever causes are within your control before the project starts
- Organize teams so knowledge is SHARED — not locked in one person's head
- Define documentation standards so anyone can pick up where someone left off
- Conduct peer reviews so more than one person understands each part
- Assign a backup person for every critical team member

2. Monitoring — What factors can we TRACK?

Monitoring is a CONTINUOUS tracking activity throughout the project. You watch for early warning signs that tell you whether a risk is becoming more likely or less likely.

■ *Analogy: Monitoring is like checking the weather forecast every day during your outdoor event planning. You are watching for signals that tell you whether to prepare for rain.*

Continuing the staff turnover example — monitor these indicators:

- General attitude and morale of team members
- How well the team has bonded and jelled together
- Interpersonal relationships among team members
- Whether compensation and benefits are causing dissatisfaction
- Availability of competing jobs in the market
- Whether the mitigation steps themselves are working

3. Management — What CONTINGENCY PLANS do we have?

Management kicks in when MITIGATION HAS FAILED and the risk has actually occurred. This is your contingency plan — your prepared response.

■ *Analogy: Management is your umbrella. Mitigation was checking the forecast. Monitoring was watching the clouds. But now it is actually raining — you open the umbrella.*

Continuing the staff turnover example — if people announce they are leaving:

- Backup people (identified during mitigation) step in immediately
- Project schedule is temporarily refocused on tasks handled by remaining staff
- Departing members spend their last weeks in 'knowledge transfer mode' — documenting everything
- New team members are brought up to speed using documentation created earlier

Student Question & Clarification

■ **Student Question: In monitoring approach, can't we say — monitor hardware when it is free to use, and once it's free use it to run tests? And for estimated resources, can we ask to buy new hardware?**

On Monitoring: That is actually part of the CONTINGENCY PLAN (what you DO when the risk hits), not the monitoring approach. Monitoring means watching for early warning signs BEFORE the risk impacts you — such as scheduling regular milestone reviews with the hardware team to get early warnings like 'hardware might be 2 weeks late' so you have time to react. Your idea of 'use it when free' is the response action, not the tracking activity.

On Buying Hardware: Buying hardware is expensive and time-consuming — procurement, setup, configuration. In a project already at risk of delay, this could worsen the problem. Pressman's contingency of SOFTWARE SIMULATION is smarter because: no procurement delay, no additional hardware budget needed, testing can begin immediately, and it is faster and cheaper to execute under pressure.

Key Exam Points

- ✓ Mitigation = avoidance/prevention — happens BEFORE the risk occurs.
- ✓ Monitoring = tracking early warning signs — happens CONTINUOUSLY throughout the project.
- ✓ Management = contingency plan — happens AFTER the risk becomes reality.
- ✓ The RMMM column in the risk table contains a POINTER/REFERENCE to the detailed plan.
- ✓ Only risks ABOVE the cutoff line get an RMMM entry.
- ✓ Mitigation is always the PREFERRED strategy — avoidance beats recovery.

Exam Questions & Model Answers

■ Exam Q: Explain the three components of RMMM with an example.

Model Answer: RMMM stands for Risk Mitigation, Monitoring and Management — three activities that together form a complete risk response strategy. Mitigation focuses on preventing the risk from occurring. For example, if high staff turnover is identified as a risk, mitigation steps include cross-training team members, assigning backups for critical roles, and improving working conditions. Monitoring involves continuously tracking indicators that show whether the risk is becoming more or less likely — such as monitoring team morale and job market conditions. Management is the contingency plan that activates when the risk actually occurs despite mitigation efforts — for example, immediately activating backup personnel, refocusing the schedule, and conducting knowledge transfer sessions with departing members.

■ Exam Q: What does the RMMM column in a risk table contain?

Model Answer: The RMMM column in a risk table contains a pointer or reference to the detailed Risk Mitigation, Monitoring and Management plan for that specific risk. It does not contain the full plan itself — rather it serves as a reference directing the reader to where the complete action plan for that risk is documented. Only risks that fall above the cutoff line in the risk table receive an RMMM entry, as these are the risks deemed serious enough to warrant a full management strategy.

TOPIC 6: Recording Risk Information — Risk Information Sheet (RIS)

What Is the Risk Information Sheet?

After identifying, projecting, and planning RMMM for each risk, you formally record everything in a Risk Information Sheet (RIS). Think of it as a detailed file card for ONE specific risk. One sheet is created per risk.

■ *Analogy: If the risk table is a summary dashboard, the Risk Information Sheet is the full detailed report behind each row.*

RIS Fields — What Goes in Each

Field	What Goes Here
Project	Name of the project this risk belongs to
Risk type	Which category: schedule, performance, cost, or support risk
Priority (1–5)	How serious is this risk overall (1=low, 5=critical)
Risk factor	A clear description of what the risk actually is
Probability	Percentage chance the risk will occur
Impact	What will happen to the project if this risk becomes real
Monitoring approach	How you will watch/track this risk throughout the project
Contingency plan	What you will do if the risk actually occurs
Estimated resources	What it will cost (people, time, money) to handle this risk

Filled RIS Example — From Your Slide 26

The following is the actual filled example shown in your slide, with explanations:

Field	Value	What It Means
Project	Embedded software for XYZ system	Which project
Risk type	Schedule risk	This risk threatens the timeline
Priority	4 (out of 5)	Fairly critical

Risk factor	Project completion depends on hardware component delivery which may be delayed	The actual risk description
Probability	60%	More likely than not to occur
Impact	Project completion delayed for each day hardware is unavailable for testing	The consequence if the risk hits
Monitoring approach	Scheduled milestone reviews with hardware group	Regular check-ins to get early warnings of hardware delay
Contingency plan	Modify testing strategy using software simulation to accommodate delay	The backup plan if hardware is not available
Estimated resources	6 additional person-months beginning in July	The cost of managing this risk

The Hardware Problem — Explained

Student Question: What exactly is the problem in the slide example? You are building SOFTWARE for the XYZ system. To TEST that software, you need a piece of HARDWARE — a physical device your software will eventually run on. The risk is: what if that hardware is not delivered on time? If hardware does not arrive → you cannot run software tests → project gets delayed. **Why not wait for hardware to be free?** Because deadlines are fixed. If your deadline is July 31 and hardware arrives July 15, you have only 2 weeks for testing but needed 6 weeks. The time is already lost — waiting does not recover it. **Why not buy new hardware?** Hardware procurement takes weeks or months — setup, configuration, compatibility checks. By the time it is ready, you have already missed the deadline. It also blows the budget. **The smart solution:** Software simulation — create a virtual/fake version of the hardware in software so testing continues WITHOUT the real hardware. Like using a driving simulator when the real car is not available.

How RIS Connects to the Full Chapter

The RIS is the final output of everything we studied in this chapter:

Step	Activity	Output
1	Risk Identification	List of all generic and product-specific risks
2	Risk Projection	Probability and impact assessed; risk table built and sorted
3	RMMM Planning	Mitigation, monitoring and management plans created
4	Recording (RIS)	All of the above formally documented in one sheet per risk

Key Exam Points

- ✓ RIS = Risk Information Sheet — ONE sheet per risk.
- ✓ It records: risk type, priority, probability, impact, monitoring approach, contingency plan, estimated resources.
- ✓ It is the formal documentation of all RMMM activity for a specific risk.
- ✓ The RMMM column in the risk table POINTS TO the corresponding RIS.
- ✓ Software simulation is preferred over hardware procurement as a contingency — faster and cheaper.

Exam Questions & Model Answers

■ Exam Q: What is a Risk Information Sheet? What does it contain?

Model Answer: A Risk Information Sheet (RIS) is a formal document used to record all relevant information about a specific identified risk in a software project. It contains the project name, risk type, priority rating, a clear description of the risk factor, its probability of occurrence, its potential impact on the project, the monitoring approach to be used, the contingency plan if the risk becomes reality, and the estimated resources required to manage it. One RIS is created per risk, and together they form the complete risk documentation for a project. The RMMM column in the risk table references the corresponding RIS for each serious risk.